

Hypothesis testing

Q1:

Café Coffee Day is concerned about the effect of the recent year long coffee advertising campaign on hot

chocolate sales. The average weekly hot chocolate sales two years ago was 984.7 pounds and the standard

deviation was 72.6 pounds. Café Coffee Day has randomly selected 30 weeks from the past year and found

average sales of 912.1 pounds.

a) State appropriate hypotheses for testing whether hot chocolate sales have decreased.

Given:

Population Mean (μ_0) 984.7

Population Standard Deviation (σ) =72.6

Sample Mean ($\bar{x}$)=912.1

Sample Size (n)= 30

Significance Level (α)=0.02

1. Forming Groups in data: Given two groups of sample and population of average weekly sales

Define null hypothesis (H_0) : Hot chocolate sales have not decreased.

$$H_0 : \mu = 984.7$$

Alternative Hypothesis (H_a): Hot chocolate sales **have** decreased.

$$H_a : \mu < 984.7$$

b) At the 2% significance level, test these hypotheses.

Sol:

1. alpha (α). $\alpha = 0.02$

Observed Value is

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}}$$

$$z = \frac{912.1 - 984.7}{72.6/\sqrt{30}}$$

$$z \approx -5.4772$$

the critical z-value for $\alpha = 0.02$ is $z_{0.02} = -2.0537$

The decision rule for a left-tailed test is to **reject H0** if $z \leq z_{crit}$.

Since the calculated test statistic ($z = -5.4772$) is less than the critical value ($z_{crit} = -2.0537$), we **reject the null hypothesis** H_0 which means chocolate sales have decreased (alternate hypothesis)

At the 2% significance level, there is **sufficient evidence** to conclude that the mean weekly hot chocolate sales have **decreased** following the coffee advertising campaign.

Q2

Suppose you start up a company that has developed a drug that is supposed to increase IQ. You know that the standard deviation of IQ in the general population is 15. You test your drug on 36 patients and obtain a mean IQ of 97.65. Using an alpha value of 0.05, is this IQ significantly different than the population mean of 100?

Solution:

Population $\sigma = 15$

Sample $n = 36$

$\bar{x} = 97.65$

Test at $\alpha = 0.05$

whether mean differs from 100 (two-sided or two tailed test).

1. State the Hypotheses:

- **Null Hypothesis (H0):** The drug has **no significant effect** (mean IQ is 100). $H_0: \mu = 100$
- **Alternative Hypothesis (Ha):** The drug has a **significant effect** (mean IQ is different from 100). $H_a: \mu \neq 100$

2. Z value

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{(97.65 - 100)}{15/\sqrt{36}} = -0.940$$

3. Critical Value: For a two-tailed test at $\alpha = 0.05$, the critical z-values are:

$$\pm z_{\alpha/2} = \pm z_{0.025} \approx \pm 1.9600$$

The decision rule for a two-tailed test is to **reject null hypothesis** H_0 if $|z| \geq z_{crit}$ (i.e., $z \leq -1.9600$ or $z \geq 1.9600$).

Since the calculated test statistic ($z=-0.9400$) is between the critical values $(-1.9600 < -0.9400 < 1.9600)$, we **fail to reject the null hypothesis** H_0 .

Using an α value of 0.05, the sample mean IQ of 97.65 is **not significantly different** from the population mean of 100

Q3

Suppose the anxiety of movies has a population that is normally distributed with a standard deviation of 1. Let's sample 40 movies from this population and obtain a mean anxiety of 94.1 and a standard deviation of 1.2039. Using an alpha value of $\alpha = 0.05$, is this observed mean significantly greater than an expected anxiety of 94?

Solution:

Population is normal with known $\sigma=1$.

Sample $n=40$

$\bar{x} = 94.1$

Test at $\alpha = 0.05$ whether mean is **greater** than 94.

1. Hypotheses

- **Null Hypothesis (H_0):** The mean anxiety is **not significantly greater** than 94. $H_0: \mu \leq 94$
- **Alternative Hypothesis (H_a):** The mean anxiety is **significantly greater** than 94. $H_a: \mu > 94$

2. Calculate (z):

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{(94.1 - 94)}{1 / \sqrt{40}} \approx -0.6325$$

3. For a right-tailed test at $\alpha=0.05$, the critical z-value is:

$$z_{crit} = z_{0.95} \approx 1.6449$$

for a two-tailed test is to **reject null hypothesis** H_0 if $|z| \geq z_{crit}$.

Since the calculated test statistic ($z=0.6325$) is less than the critical value ($z_{crit}=1.6449$), we **fail to reject the null hypothesis** H_0

Using an α value of 0.05, the observed mean anxiety of 94.1 is **not significantly greater** than the expected anxiety of 94.

Q4

A cereal brand claims that its boxes contain at least 450 g. A consumer group samples $n = 64$ boxes, finds mean to be 447 g. Population standard deviation $\sigma=8$ g. Test at $\alpha=0.02$ whether the boxes contain less than claimed.

Solution:

Claim $\mu \geq 450$

Sample $n=64$

$\bar{x} = 447$,

population $\sigma=8$

Test at $\alpha=0.02$ whether boxes contain **less** than claimed

Hypotheses:

- **Null Hypothesis (H0):** The boxes contain 450 g. $H_0: \mu=450$
- **Alternative Hypothesis (Ha):** The boxes contain **less than** 450 g. $H_a: \mu < 450$

2. Calculate the Test Statistic (z):

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{(447 - 450)}{8 / \sqrt{64}} = -3.0$$

Critical Value: For a left-tailed test at $\alpha=0.02$, the critical z-value is:

$$z_{crit} = z_{0.02} \approx -2.0537$$

- for a left-tailed test is to **reject H0** if $z \leq z_{crit}$

Since the calculated test statistic ($z=-3.0000$) is less than the critical value ($z_{crit}=-2.0537$), we **reject the null hypothesis H0**.

So at the $\alpha=0.02$ significance level, we can conclude that the cereal boxes contain **less than** the claimed 450 g

Q5

A researcher wants to know whether exposure to paid political ads affects knowledge about a particular issue. In the general population (who know nothing about the issue), the mean score on a multiple-choice test is 40 with a standard deviation of 5. Five individuals who had seen the ads scored: 48, 41, 40, 51, and 50. At the 0.05 level of significance, use a z-test to determine whether people who see the ads score differently on this test.

Solution:

Population mean (no prior knowledge) $\mu_0=40$

population $\sigma=5$

Five people who saw ads scored: 48, 41, 40, 51, 50.

Use a **z-test** at $\alpha=0.05$ to see if scores differ.

1. Calculate sample mean $\bar{x}$

$$\bar{x} = \frac{48+41+40+51+50}{5} = 46.0$$

2. Hypotheses

- **Null Hypothesis (H_0):** The mean score for people who see ads is **no different** from the general population. $H_0: \mu = 40$
- **Alternative Hypothesis (H_a):** The mean score for people who see ads is **different** from the general population. $H_a: \mu \neq 40$

2. Z value

$$z = \frac{\bar{x} - \mu_0}{\sigma / \sqrt{n}} = \frac{(46 - 40)}{5 / \sqrt{5}} \approx -2.6833$$

For a two-tailed test at $\alpha=0.05$, the critical z-values are:

$$\pm z_{\alpha/2} = \pm z_{0.025} \approx \pm 1.9600$$

5. Make a Decision: The decision rule for a two-tailed test is to **reject** H_0 if $|z| \geq z_{crit}$ (i.e., $z \leq -1.9600$ or $z \geq 1.9600$).

Since the calculated test statistic ($z=2.6833$) is greater than the critical value ($z_{crit}=1.9600$), we **reject the null hypothesis** H_0 .

At the 0.05 level of significance, there is **sufficient evidence** to determine that people who see the paid political ads score **differently** on the test.
